Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping AI products end to end, from **LLM** workflows and Python and Node.js **APIs** to **React/Next.js** UIs, plus internal tools that removed manual work, building daily with Claude Code.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant **AI education platform** on Azure SQL and Blob Storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through query tuning.
- Rebuilt **LLM orchestration** with FastAPI background workers, explicit job states, and failure isolation, tripling throughput across 1K+ AI jobs while integrations ran 5K+ monthly jobs at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped an AI grading product end to end with **React** views and Flask APIs, chaining GPT-4o, Claude, and Qwen-VL for rubrics, OCR, and feedback for 150+ instructors and 10K+ daily requests.
- Cut instructor feedback turnaround 60% with concurrent batch processing of 1K+ files, timeout-aware model fallback, and per-call **cost tracking**, deployed through GitHub Actions to Render and Vercel.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated cross-platform data-sharing **workflows** on a microservice architecture that eliminated 35% of manual data entry, partnering with Microsoft and GitHub stakeholders from requirement analysis through deployment in Agile sprints.
- Optimized large-scale API integrations with pagination, field filtering, and caching, improving response times 30%, and led an ETL redesign on AWS Glue and S3 that boosted **pipeline efficiency** 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **internal tools** for sales and operations teams, a TypeScript and React CRM with Node.js services for leads, billing, resource allocation, and notifications, syncing 2M+ records in real time.
- Used Google Analytics on feature usage and drop-off points to drive product changes that cut resource wastage 42% and raised lead conversions 15%, shipping via Jenkins **CI/CD** at 95%+ build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Gathered client requirements, designed **UI/UX** flows in Figma, and presented the end-to-end architecture, reducing design errors 25% before building Python Flask APIs for bookings, pricing, and coupons.
- Automated refund, dispute, and chargeback handling through idempotent payment **webhooks**, cutting manual refund handling 70%, and synced booking status in real time through Firebase across 50K+ records.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML, CSS
Frontend: React, Next.js, TailwindCSS, Shadcn, Responsive Design, UI/UX Design, Figma
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks
AI: Claude, GPT-4o, LLM Orchestration, Prompt Chaining, RAG, Embeddings, Claude Code
Data & Infra: PostgreSQL, Supabase, MongoDB, Firebase, Stripe, AWS, Vercel, Docker, CI/CD
Practices: Internal Tools, Workflow Automation, Rapid Prototyping, Cross-functional Collaboration

Projects

AI Customer Interview Agent | *Next.js, TypeScript, FastAPI, Claude, PostgreSQL, TailwindCSS*

- Designed a Next.js and FastAPI app where an **LLM agent** runs adaptive customer interviews, extracts structured themes with citations to responses, and surfaces insights in an animated dashboard for product teams.

Big-Brain Live | *LLaMA-2, Convex, Clerk, Shadcn, FAISS*

- Built and deployed a collaborative **AI** document app with LLaMA-2 RAG, Convex real-time sync, Clerk auth, and a polished Shadcn UI, cutting document review time 60% for 200+ users.

PiXMod Live | *Next.js, Stripe, SAM, Stable Diffusion, CLIP, MongoDB*

- Built and launched an AI SaaS product in Next.js with SAM segmentation, Stable Diffusion inpainting, CLIP search, and **Stripe** subscriptions, lifting premium conversion 18% while processing 1K+ edited assets.